Journal of Retailing and Consumer Services 85 (2025) 104313

Contents lists available at ScienceDirect

Journal of Retailing and Consumer Services

journal homepage: www.elsevier.com/locate/jretconser

Unlocking the potential of NFTs in branding: An exploration of NFT-based brand experience

Wenjie Li *, Graciela Corral de Zubielqui, Sally Rao Hill

Adelaide Business School, South Australia, Australia

ARTICLE INFO

Keywords:
Non-fungible tokens
Brand experience
Digital branding

ABSTRACT

This study explores the evolving intersection of branding and digital assets through the lens of non-fungible tokens (NFTs), focusing on their role in shaping dynamic brand experiences. We propose a typology framework that examines how NFTs contribute to brand experience design and provides their implications for brand-consumer relationships. The research analyses five distinct NFT functions—storytelling media, identity badges, product access pass, change medallion and gamification element—and connects these roles to five types of brand experience design: brand heritage, community, product orientation, collaboration, and gamification. The findings contribute to digital branding literature by advancing the understanding of the function of digital assets within the brand experience design. This study offers a structured understanding of the value of NFTs in digital brand building by providing the roles NFTs play in brand experience. It explores the dynamic potential of brands to integrate NFTs into their strategies in the evolving Web3 environment. Finally, the industry pattern identified in this study provides insights for scholars and practitioners seeking to utilise NFTs effectively.

1. Introduction

Blockchain enables the creation and management of digital assets such as NFTs (Non-Fungible Tokens), which are integral to the metaverse, representing ownership of virtual land, items, and experiences (Chalmers et al., 2022). The metaverse based on blockchain has been described as a new iteration of the internet - a new paradigm for interacting with digital technology in immersive virtual environments (Dwivedi et al., 2022). The Software Policy and Research Institute (SPRI, 2021) predicts that the metaverse will continue to encompass more economic activity and be driven by metaverse device usage, digital human development, brand alliances, and NFTs. Among them, the potential of NFTs in the metaverse is significant (Lee et al., 2023). A report shows that in 2023 the global NFT market was valued at approximately $26.9 billion, with projections estimating its market size to reach $147.24 billion by 2026 (View, 2024). In addition to their economic functionality, NFTs have become an important tool for decision-makers in marketing operations in the context of the metaverse (Barrera and Shah, 2023).

NFTs are encrypted assets on the blockchain with unique identification codes and metadata that can distinguish them from each other (Yilmaz et al., 2023). As a digital asset, NFTs are different from other cryptocurrencies because they are inherently not transmittable, making the ownership and trading of digital assets technologically possible (Chalmers et al., 2022), which brings great opportunities to brand marketing (Zhang and Phang, 2024; Hofstetter et al., 2022; Xie et al., 2023). Brouard (2024) reveals the multifaceted nature of NFTs, from keys to access exclusive privileges to standalone products whose value is intrinsically tied to market dynamics, highlighting the potential transformative impact of NFTs on consumer experience. Alexander and Bellandi (2022) found that NFTs are associated with uniqueness, prestige, and authenticity, which reshape the value creation in brand marketing. Due to the inherent characteristics of NFTs, which enable them to assume multiple roles in marketing, NFTs have great potential to support a brand's continuous development in the metaverse by creating dynamic experiences (Colicev, 2023).

Yet, while the potential of NFT in branding has been recognised, there has been little exploration of how brands use NFT in their marketing strategies (Lee et al., 2023; Chen et al., 2025). Recent literature has increasingly emphasised the urgency of studying different NFT strategies brands adopt (Cho et al., 2024b; Xie et al., 2024; Boukis, 2024). Brouard (2024) noted that it is crucial to address this gap to

* Corresponding author. Room 911, 12/10 Pulteney St, Adelaide, SA, 5005, Australia.
E-mail addresses: wenjie.li01@adelaide.edu.au (W. Li), graciela.corraldezubielqui@adelaide.edu.au (G. Corral de Zubielqui), sally.raohill@adelaide.edu.au (S. Rao Hill).

https://doi.org/10.1016/j.jretconser.2025.104313
Received 21 January 2025; Received in revised form 17 March 2025; Accepted 27 April 2025
Available online 3 May 2025
0969-6989/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

understand further how NFTs influence consumer behaviour in various ways. Furthermore, the demand to investigate how various NFTs align with the expectations of distinct consumer groups highlights the need to comprehensively examine diverse brand NFT strategies (Chen et al., 2025). Also, the researchers emphasise understanding the impact of different types of NFT on reward programmes (Boukis, 2024), consumer behaviour (Chen et al., 2025), brand-consumer interaction (Kim et al., 2025) and brand-related consequences (Peres et al., 2023), which prioritise the need for a structured categorisation of values NFT provides for brands. The lack of a comprehensive understanding of the roles of NFT in brand strategies further limits the strategic value of NFTs in digital branding. In addition, the multifaceted characteristics of NFT highlight the importance to observe how different NFT roles are deployed across various sectors, which further allows for a targeted strategy alignment to leverage NFT in branding (Zhang and Phang, 2024; Murtas et al., 2024). In summary, the insufficient exploration of the varied roles of NFTs, sector-specific differences, and the design of NFT-based brand experiences limit a further understanding of the impact of brand strategies that incorporate NFT in Web3. A typology of NFT-based brand experience design is crucial to provide a framework to analyse how brands strategically deploy NFT to cater the evolving market in the metaverse and achieve marketing objectives (Bailey, 1994; Ortiz-Avram et al., 2024). To provide valuable insights into how brands foster innovative NFT-based strategies in a virtual landscape, this study proposes a framework of the strategic application of NFT in brand dynamic experience design. Specifically, our research questions are as follows:

1) How can the key characteristics of brand NFTs be used to develop a comprehensive typology that classifies dynamic brand experiences?
2) Is there an industrial pattern in adopting specific brand experience types, and what distinct roles do NFTs serve within these experiences?

In addressing these questions, we focus on the 118 NFT collections released by 85 brands. By analysing the descriptions associated with these collections based on liminoid theory, we develop a typology of NFT-based brand experience to understand better the NFT field's transitional nature and trending phenomena. Based on the findings of typological analysis, we provide a comprehensive framework on how NFTs shape brand experience, contributing directly to the call for detailed exploration of brand NFT strategy highlighted in the digital branding literature. The identification of NFT roles in brand experience fills the gap of how NFTs influence brand strategy deployment in the context of Web3. We also contribute to marketing practice. Marketers can better understand how to integrate NFT into their brand strategies. This study reveals how NFTs influence brand strategy evolution across industries by illustrating industry patterns of brand experiences. As an exploration of brand NFT applications, this research has the potential to bridge the gap between academia and the marketing industry. It provides valuable insights to academic researchers while enabling marketers to make informed decisions and develop effective strategies in the evolving world of NFT branding.

This paper is organised as follows. First, we introduce the theoretical background to establish a solid foundation for the analysis in this study. Then, we frame the literature review to illustrate the related context. We proceed with the methodology, detailing its design and each step in its application. After that, we illustrate the findings and present the resulting framework. Finally, we present propositions drawn from our findings in the discussion section, followed by a summary of the study's contributions in conclusion.

2. Theoretical background and literature review

2.1. Liminoid theory and consumers' virtual experiences

The conception of the metaverse as an integration of physical and virtual worlds emphasises its immersive and interactive nature, facilitating a temporary suspension of traditional norms and boundaries (Park and Lim, 2023; Yoo et al., 2023). It is believed that brands are able to communicate value differently in the transitory metaverse compared to other digital platforms (Yoo et al., 2023; Park and Lim, 2023). As an important part of the metaverse (Cohen and Fishbain, 2025), NFTs represent digital art, virtual real estate, or interactive experiences (Kumar et al., 2025). Building on the understanding that NFTs offer unique and transformative brand experiences that bridge digital and physical interaction (Kumar et al., 2025; Joy et al., 2022), liminoid theory with a focus on examining transformative experience provides a compelling lens for exploring NFT-based brand experience. According to the work of Darveau and Cheikh-Ammar (2021), the fluid value of NFTs associated with digital and physical transitional experiences and spaces can be considered extrinsically liminal products. Additionally, NFTs also present intrinsic liminality nature as they function simultaneously as ownership, authenticity and access (Darveau and Cheikh-Ammar, 2021). The combination of intrinsic and extrinsic liminality in NFTs makes liminoid theory a suitable framework for analysing brand experience designs that incorporate NFTs as a strategic tool. Liminoid theory was originally tied to traditional societies' structured phases of rites of passage (Taheri et al., 2017). Turner (1983) then extended the concept of liminal spaces into voluntary, individual-based transitions that allow for play and creativity, making the concept applicable to a modern setting. Turner (1983) concepts of liminality and ritual processes have been applied in the social sciences to theorise the physical, psychological, and communal spaces that facilitate experimentation and social identity transformation (Laver et al., 2022; Taheri et al., 2016; Tan, 2024). In a recent study, the liminoid theory has shown its relevance in the virtual context (Jung and Pawlowski, 2014), which highlights the adaptability of liminoid theory to interpret changing forms of consumption and identity within both traditional and modern settings. Jung and Pawlowski (2014) propose the virtual liminoid theory, which hypothesises that the virtual environment is suitable for liminoid activities because it separates users from the real world in space and time. In contrast, both Technology Acceptance Model (TAM) and Self-Determination Theory (SDT) primarily address consumers' preferences and functional motivations rather than the experiential design perspective (Davis, 1985; Gagné and Deci, 2005), making liminoid theory an appropriate lens for providing a comprehensive understanding of how brands use NFTs to craft complex, transitional brand experiences.

Liminoid theory reveals new possibilities for engagement within virtual spaces and serves as a valuable framework for understanding transformational dynamics in contemporary marketing contexts (Taheri et al., 2017; Cappellini and Yen, 2016). In marketing, liminoid is considered a suitable perspective to explain experiential consumption in post-industrial hedonism (Taheri et al., 2016). Turner (1998) also explicitly links the liminoid experience to the consumer experience. Besides, scholars suggest that marketing and consumption symbolically draw on aspects of ritual processes to utilise the potential of liminoid theory for change and transformation (Hackley et al., 2021; Park and Lim, 2023). Liminoid theory's experiential, anti-structural, and transitional dimensions (Turner, 1998) align with brands' efforts to create unique, immersive, and enjoyable experiences through NFTs (Xie et al., 2024). Moreover, its growing application in consumer and marketing research highlights its relevance in capturing the evolving nature of brand-consumer relationships in a rapidly changing marketplace (Leigh et al., 2006; Scott et al., 2017; Darveau and Cheikh-Ammar, 2021). Liminoid theory provides foundational insights into how brands can design transformative NFT-based experiences, consistent with the

theory's focus on the liminal space of play and the goal of meaningful brand experiences. Therefore, in this study, liminoid theory functions as a contextual perspective to understand the transitional nature of NFT-based brand experiences and as a guiding framework for content analysis to provide dimensions to code and interpret these experiences. The core attributes of temporary escape, playfulness, and ritual elements embedded in liminoid theory provide a valuable lens for this study, facilitating a deeper understanding of the brands' design of the NFT collection to introduce a dynamic brand experience. Table 1 illustrates the application of liminoid theory in this study.

By leveraging NFTs, brands can create liminoid spaces where consumers engage in blockchain-based experiences marked by authenticity and exclusivity. Temporary escape in liminoid theory represents a phenomenon that subverts the quotidian structure (Turner, 1983). McKercher and Lui (2014) suggest that liminal spaces and experiences allow individuals to set aside conventional standards amidst relative anonymity and freedom. The anti-structural nature of liminoid experiences driven by NFT facilitates brands to encourage consumers to escape the routines of daily life and conventional brand interactions. Therefore, the motivation to escape, central to the liminoid theory, resonates with how brands use NFTs to initiate new brand experiences.

Playfulness in liminoid theory represents an attribute that breaks down rigid social interactivity (Taheri et al., 2017), which appears to have implications for the design of interactions in brand NFT experiences. Turner (1983) emphasised that liminoid spaces encourage participants to explore new identities and engage in playful interactions not bound by traditional roles. The virtual attributes of NFTs provide consumers with brand interaction beyond physical space (Ante, 2023). By incorporating NFTs into experiences, brands are able to create immersive environments that allow consumers to actively participate in, collect, trade, or use exclusive digital assets. Moreover, the ownership and uniqueness inherent in NFTs add unprecedented playfulness, consistent with the emphasis on exploring identity in liminoid theory (Jung and Pawlowski, 2014; Hackley et al., 2021).

As Turner (1983) described, ritualistic elements involve symbolic markers that signify important transitions or achievements in a liminal experience. In the context of brand NFT experiences design, the rewards obtained in brand NFT experiences, represented by digital collectibles, are a motivational mechanism and represent the transformative process of participation and co-creation, marking significant transitions (Bilgihan et al., 2024; Lee et al., 2024a).

In summary, NFTs bridge the virtual and real worlds, offering dynamic experiences that encourage ongoing consumer interaction with brands and the evolution of brand relationships. This dynamic nature aligns with the concept of liminoid experiences. Therefore, liminoid theory provides an analytical framework for understanding how brands can use NFTs to create dynamic consumer experiences. This approach allows us to explore how brands can create meaningful consumer interactions. The motivation for escape is closely linked to the playfulness and ritualistic elements (Taheri et al., 2016), offering a more holistic understanding of how NFTs shape transformative and dynamic brand experiences. By applying liminoid theory, we aim to provide a structured framework to understand better how NFTs can guide the holistic picture of brand NFT experience design and the transfer of brand meaning facilitated by the dynamic experience.

Table 1
Conceptual connection between liminoid theory and brand NFT experience design.

Attribute	Application in liminoid theory	Connection to brand NFT experience	Analytical category
Temporary escape	Turner (1983) conceptualises liminoid experiences as anti-structural phenomena that subvert the quotidian and prestigious structures (Daskalaki and Simosi, 2018).	NFTs allow consumers to escape from their daily lives and existing brand interactions and enter a new space that disrupts the traditional brand experience.	Purpose
Playful	Liminoid experiences are oriented to playfulness (Turner, 1998; Taheri et al., 2016).	The interactive attribute of NFTs enables brands to create immersive engagement experiences that encourage exploration and innovation within the brand ecosystem.	Interactivity
Ritualistic element	The symbols mark important transitions (Turner, 1983; Darveau and Cheikh-Ammar, 2021).	The rewards gained throughout brand experiences serve as ritualistic elements, representing the transformation of the significant brand experience.	Rewards

2.2. NFT in brand experience

Blockchain provides a secure ledger for digital assets stored decentralised, thereby reducing transaction costs and improving data security (Wang et al., 2021). The main feature of blockchain technology, data accuracy and security, make blockchain technologies highly relevant for the evolution of marketing strategies (Rejeb et al., 2020). Among them, NFTs have brought a great impact on the brand's strategic design, interaction mechanism, and other marketing aspects (Chohan and Paschen, 2023). The traceable and assets nature of NFT highlights its potential in achieving both economic and noneconomic goals (Chohan, 2021; Chandra, 2022). Despite research highlighting the potential challenges that NFTs may pose to consumer protection in terms of finance (Gupta et al., 2022) and privacy (Das et al., 2022), NFTs change the way businesses manage their brands to capture consumer interest (Kshetri, 2022; Kaur et al., 2024). The integration of blockchain and NFTs into digital branding represents a transformative shift in how brands manage engagement (Lee et al., 2023), enhance experience (Park and Lim, 2023) and build brand equity (Cho et al., 2024a).

Recent literature about NFT in branding only started in 2022 and focuses on three themes: NFT characteristics and brand outcomes, roles of NFTs in brand marketing, and factors influencing consumer attitude toward brand NFTs, as shown in Table 2.

Studies in NFT characteristics and brand outcomes group assess the connections between NFT intrinsic features and branding strategies, which provide insights into how NFT's unique characteristics, such as authenticity and scarcity, reshape consumer–brand interactions (Lee et al., 2023; Ramadan, 2024). In the exploration of NFT roles in brand marketing, studies identify NFT as a promotional tool (Cho et al., 2024b), communication tool (Kim et al., 2025), reward scheme (Boukis, 2024) and distribution technology (Hofstetter et al., 2022). Lastly, the theme of factors influencing consumer attitude toward brand NFTs focuses on the antecedents influencing consumer perception and behaviour in relations to brand NFTs.

As discussed, NFTs have greatly impacted the brand's strategy design and consumer behaviour (Chohan and Paschen, 2023). Brand experience provides a valuable perspective for studying the typology of NFTs because it focuses on the experiential dimension that connects the functional and emotional connection between brands and consumers. In digital branding, brand experience has been shown to be directly linked to multiple dimensions of brand equity (Hepola et al., 2017). Furthermore, Prentice et al. (2019)'s work illustrates that brand experience directly and indirectly impacts customer engagement. Based on the constructs of brand experience (Brakus et al., 2009), we believe that NFT impacts brand experience in five aspects: sensory, affective, cognitive, behavioural and social. First, studies have confirmed that NFTs have great potential in displaying visual aesthetic features (Chen et al., 2023),

Table 2
Summary of literature on NFT and branding.

Themes	Publications	Method	Key findings
NFT characteristics and brand consequences	Lee et al. (2023)	Quantitative	Brand NFT attributes include scarcity, financial value, prestige, uniqueness, originality, and communication consistency, promote favourable brand attitudes, which in turn enhance brand commitment, brand NFT purchase intention, and positive interactions with BNFTs.
	Zhang and Phang (2024)	Quantitative	The scarcity, exclusivity, design aesthetics, and novelty of NFTs have a positive impact on consumers' perceived hedonic value, further contributing to their emotional attachment to luxury fashion brands.
	Ramadan (2024)	Qualitative	NFTs contain unique information related to a user's identity, findings reveal a SBT-led cycle.
	Cho et al. (2024a)	Mixed	NFT's key features – scarcity, resaleability can create consumer equity can translate into purchase intent.
	Xie and Muralidharan (2023)	Quantitative	NFT users prefer branded NFTs that offer exclusivity and financial benefits. These values enhance the branded NFT experience, which in turn increases the willingness to purchase NFTs and build brand loyalty.
	Xie et al. (2024)	Quantitative	Value perception of brand NFT (unique, entertainment, and expressive value) contributes to the individual's sense of status consumption and personality innovativeness, which further leads to positive brand WOM.
Roles of NFT in brand marketing	Hakkarainen and Colicev (2023)	Qualitative	NFT as one of the blockchain-enabled advances can be utilised by brands to roll out technology-focused service, customer-focused service, and product innovations
	Colicev (2023)	Conceptual	From a branding perspective, NFTs can be viewed as both a brand component as well as an independent brand asset, showing potential for connecting brands and consumers.
	Mao et al. (2024)	Quantitative	The stimuli in digital fashion impact consumers' attitudes, satisfaction, involvement.
	Cho et al. (2024b)	Quantitative	The scarcity, authenticity, and trendiness of NFTs have a positive impact on consumer behaviour. Therefore, NFT is useful as a promotional tool aimed at building brands and cultivating purchase intentions.
	Hofstetter et al. (2022)	Conceptual	NFT challenge established marketing understanding of digital ownership, uniqueness, and value; authenticity, status, and sharing; and branding and distribution.
	Boukis (2024)	Quantitative	The tokenised rewards (NFT) enhance the attractiveness of reward schemes for customers.
Factors influencing consumer attitude toward brand NFTs	Chen et al. (2025)	Quantitative	Three distinct clusters of brands NFT consumer were identified, namely speculators, casual collectors, and crypto natives, as well as their characteristics and behaviours related to obtaining brand NFT.
	Wu et al. (2024)	Quantitative	Risk perception, trust and attitude were significantly and positively related to participants' intention to purchase NFTs.
	Kim et al. (2025)	Quantitative	Consumers prefer visualised NFTs over non-visualised ones, with increased trust in visualised NFTs.
	Prasad et al. (2023)	Quantitative	Perceived benefits (including enjoyment, exclusivity, and value appreciation) are the most important determinants of NFT purchase intention.
	Lee et al. (2024b)	Qualitative	The economic, functional (uniqueness), and experiential (self-directed pleasure and affiliation) value of virtual luxury NFT wearables directly influences consumer purchase intent.
	Eastman et al. (2024)	Qualitative	Three subsegments were identified: Pessimists, Futurists, and Authenticators to explore young people's perceptions of NFTs. Pessimists view NFTs as risky and fleeting, futurists see them as the future of investment, and authenticators perceive them as prestigious digital assets.

providing novel aesthetic experiences (Reshef Kera et al., 2023). Then, NFTs establish affective connections between brands and consumers (Zhang and Li, 2023; Mostafa and Kasamani, 2021). The brand NFT experience has the potential to shape consumer emotions by fostering a shared vision between the brand and the individual. This, in turn, strengthens the emotional connection and identification between the consumer and the brand (Van der Westhuizen, 2018). Third, by serving as brand components, NFTs facilitate the creation of authentic and creative brand stories and values that encourage consumers to explore the brand's identity (Colicev, 2023). As NFT blends online and offline experiences by serving as multi-role tools, a growing body of literature highlights NFTs' critical role in brand experience strategies (Xie et al., 2023; Lee et al., 2023).

NFTs also impact consumers' behavioural responses (Lee et al., 2023). By facilitating interactive activities such as collecting, participating in exclusive events, and co-creating, NFTs help enhance consumer engagement with brands and expand the scope of brand-consumer engagement across digital and physical touchpoints (Bao et al., 2024; Colicev, 2023; Cho et al., 2024a; Lee et al., 2023). Finally, the social interactions surrounding brand NFTs provide entertainment that brands can leverage to develop pervasive and powerful communities. Colicev (2023) believes that NFTs can play a role in the entire marketing channel as an independent brand component and create value for the brand by building an NFT brand community.

Despite these advancements, there remain substantial gaps in our understanding of how digital assets enable the dynamism and potential for brand experience. Different brand experiences have received significant attention in the digital environment for their unique impact on consumer perceptions and behaviours (Ong et al., 2018). This gap highlights a critical need for new frameworks and typologies that address these evolving consumer-brand interactions. Here, NFTs as brand assets present an opportunity to meet the demand for brand experiences that are both customisable and scalable across different digital contexts (Bilgihan et al., 2024). Therefore, recognising the different types of brand experience is critical to understanding how brands can design more effective and engaging marketing strategies in the context of evolving technology (Paik and Lee, 2021). Moreover, existing research on digital brand experience highlights the importance of multi-channel strategies which engage consumers across physical and digital channels to foster cohesive brand interactions (Rodríguez-Torrico et al., 2020; Massi et al., 2023), presenting an opportunity to examine how such strategies address evolving consumer expectations for accessibility of brand experience. Beyond their role in integrating multiple channels to deliver better brand experiences, the roles of NFTs encompass a broader range of strategic functions (Colicev, 2023). While the existing literature explores the potential of NFTs in branding (Colicev, 2023), there is a lack of comprehensive understanding that categorises the multifaceted roles of NFTs, which limits the exploration of strategic value of NFTs in digital branding, particularly regarding its impact on the consumer-brand relationship (Park and Lim, 2023). It is also important to note that due to the multiple underlying characteristics of NFTs (Zhang and Phang, 2024) and the diverse objectives of industries in Web3 (Murtas et al., 2024; Chicotsky, 2023; Florès and Lejealle, 2023), focusing on sector-specific NFT deployment differences allows for a targeted strategy alignment to leveraging NFTs in branding.

By studying brand NFT experience design, a typology can be developed to classify NFTs based on their unique marketing attributes in combination with consumer-brand interactions, providing a comprehensive way to understand the application of NFTs of the brand. A typology of NFTs in branding would contribute significantly to the three emerging themes. First, it qualitatively explores the interpretative framework further to assess the impact of different NFTs on branding consequences (Chen et al., 2025). Second, the typology of NFTs facilitates insights into the various applications of NFTs in branding strategies (Xie et al., 2024). Additionally, the typology of NFT enriches the exploration of consumer drivers by linking different types of NFTs with the diverse reward designs from brands (Boukis, 2024). Finally, existing investigations into the industrial context of NFT branding have predominantly focused on the luxury fashion industry (Zhang and Phang, 2024; Kim et al., 2025). However, the literature highlights the need for a broader understanding of NFT design employed by brands across diverse sectors(Cho et al., 2024a; Chen et al., 2025).

3. Method

This study employs a mixed-method design to develop a typology of a dynamic brand experience based on NFTs. Specifically, the study follows a configuration approach to develop insights.

3.1. Sample

Promoting brand NFTs experience involves explicitly describing the brand stimulus designed in NFTs and what experience consumers can expect (Lee et al., 2023). To examine the different brand NFT experiences, we analysed the textual description of brand NFT collections published online. An NFT collection is a unique group of NFTs that share common features and are often distinguishable (Mekacher et al., 2022). We chose the brand NFT collection as an analysis unit because it provides a meticulous and comparable description of the intended brand experience designed by the brand. These descriptions are crafted because textual data is relevant and necessary source material to answer questions of interest and can be used to describe and explain organisational behaviour, values, and norms (Alexa, 1997). In this study, the brand NFT collection descriptions are meaningful in illustrating the differences in the dynamics of brands' virtual experiences empowered by NFTs.

Adopting the methodological approach described by Bao et al. (2024), we initially searched for the keywords "brand", "NFT", "Non--fungible token", and "branded NFT" on Google. As Google's ranking algorithms prioritise relevant content, we read the first 100 results to understand the scope better. While general search engines like Google may introduce bias due to inherent personalised algorithms, it is an effective tool for obtaining preliminary information (Stansfield et al., 2016; Piasecki et al., 2018). Then, a cross-reference snowballing method (Wohlin, 2014) was applied to expand target cases. This process involved systematically reviewing social media, news websites, blogs, and others until all cases in the research scope were included. To enhance the relevance of the sample to our research objectives, we established a set of specific inclusion criteria. First, brands for which NFTs serve as the main product were excluded from the data set to focus on brands that integrate NFTs as part of their marketing strategies. Second, we selected brands recognised on the Forbes lists, which rank publicly traded companies based on composite scores, especially market interest related data (Forbes, 2024). Forbes lists are widely used by researchers in branding studies (Getchell and Beitelspacher, 2020; Adegbola et al., 2018). This step improves the data's relevance by capturing the contemporary market recognition and influence. Then, according to the triangulation principle (Yin, 2004; Mishra et al., 2022), data from more than three sources (brand website, social media, NFT marketplaces) were cross-verified to enhance data validity. Finally, the descriptions of the NFT of the brand used for consequence analysis were sourced from official brand websites and verified brand accounts on social media and NFT marketplaces. Throughout the research period, we added newly launched cases that met the inclusion criteria to the list. This process ran from July 2024 to October 2024, resulting in 118 collections from 85 brands (see Appendix A).

3.2. Data analysis

Configuration has been applied both practically in organisation management (Harms et al., 2009) and academic research as an

analytical method to identify patterns and profiles (Flynn et al., 2010), both of which are grounded in configuration theory (Miller, 1996). We followed a configuration approach (Flynn et al., 2010; Mair et al., 2012) to explore the brand NFT dynamic experience. Configuration is an analytical method that configures and combines variables that have proven useful in classifying disordered phenomena into relatively similar groups (Fiss, 2007). In this study, the configuration method supports interpreting the distinct features across the clusters identified, allowing variables to be organised into a coherent and meaningful pattern (Flynn et al., 2010) to enhance our understanding of brand-consumer interactions in digital contexts. This approach is influential in ensuring the effectiveness of organisational strategies and opposes the assumption of "one best way to organise" (Sarason and Tegarden, 2003). Depending on the configuration perspectives, the organisation's different strategic environments and structures may lead to different characteristics and results (Fiss, 2007). It is widely used in marketing literature to develop typologies (Vorhies and Morgan, 2003). The configuration approach is useful for capturing the complexity of brand NFT-based experiences as it allowes for the discovery of emerging patterns (Miller, 1996; Flynn et al., 2010). By adopting a configuration approach, this study explores how different elements of brand NFT experiences interconnect, providing a structured but flexible analysis of a typology of NFT-based brand experiences.

Our analysis proceeded in two steps. First, using content analysis, we generated codes of purpose, interactivity and rewards from NFT collection descriptions. Content analysis provides a structured method for collecting and analysing unstructured qualitative data from various sources such as interviews, social media, or websites (Morgan, 1993; Snelson, 2016). Given that NFTs represent a new phenomenon in the context of branding, we employed a hybrid approach to conduct content analysis, where deductive analysis first identifies codes based on predefined conceptualisations derived from liminoid theory, followed by inductive grouping to allow broader categories to emerge, enhancing the analysis with structure and discovery. Hybrid content analysis provides flexibility in analysis, allowing researchers to explore complex phenomena (O'Connor and Joffe, 2020; Elo and Kyngäs, 2008). Therefore, we adopted content analysis to extract key elements corresponding to the three conceptualisation aspects as a basis for cluster analysis. Building on the content analysis, we then applied cluster analysis to identify distinct types of brand NFT experiences. Cluster analysis refers to classification procedures that uncover homogeneous groups underlying a data set (Aldenderfer and Blashfield, 1984; Henry et al., 2015). Cluster analysis has been widely employed in marketing research to detect similarities and differences within groupings and derive a typology (Kau et al., 2003). Therefore, cluster analysis was deemed suitable for addressing under-theorised phenomena and organising brand NFTs into groups suggested by the data. The purpose of clustering in this study based on content analysis is to gain insight into the similarities in brand NFT experience design.

The content analysis method was applied through a systematic, collaborative process involving three coders. The primary coder thoroughly read the original text of brand NFT descriptions and performed text recognition on the data based on purpose, interactivity and rewards. The preliminary codes were then discussed with two other coders to ensure mutual understanding of the codes. Through this step, our team identified terms from the original text for each conceptualisation using in vivo codes (words taken from the texts), assigning codes that align with each predefined conceptualisation (themes) (Elo and Kyngäs, 2008). The terms taken from the text formed the basis for generating themes of purposes, interactions and rewards. Second, we listed the codes and grouped them into broader, more abstract conceptual high-level themes, which gave us a deeper understanding of previously unexplored aspects such as NFT phenomena (Elo and Kyngäs, 2008). The authors discussed the identified terms, consolidating those with theoretical similarities and grouping them into higher abstracted themes. When creating codes, the authors returned to the text to check whether the first-order codes fit into higher abstract themes. After refining the codes, all three coders independently coded a subset of the data. Intercoder reliability was then assessed to ensure consistency and validity in the coding process (Burla et al., 2008). Fig. 1 shows the coding process along with examples for codes of purposes, interactivities and rewards. We employed Krippendorff's alpha as a reliability measure to mitigate the limitations with manual content analysis. Alpha values recorded above 0.67 are considered reliable (Hayes and Krippendorff, 2007). In our study, Krippendorff's alpha value for three coders was 0.79, indicating the reliability of the data.

The categories resulting from content analysis were used to create binary vectors of categories (using 0 and 1 to represent either a theme absent or present). After deleting attributes with less distribution (accounting for less than 3 %), 19 themes were identified to characterise brand NFT experiences for cluster analysis. Chi-square tests of independence were conducted to evaluate the significance of the categorical themes. Chi-square tests revealed that all themes were significantly different between groups (all p-values <0.05), indicating that each variable contributes meaningfully to the distinction between clusters.

Hierarchical cluster analysis uses the Ward Jr (1963) method to generate a dendrogram (or tree diagram) to understand the appropriate number of clusters. Hierarchical clustering starts with each category as one cluster, and pairs of clusters are merged as one moves up the hierarchy (Murtagh and Contreras, 2012). The dissimilarity measure for different groups was achieved by calculating the Minimum Increment of the Sum of Squares (MISSQ) (Ward Jr, 1963). Hierarchical cluster analysis is able to achieve our research goal, as it does not provide predefined clustering solutions and it recognises clusters by identifying similarities in distances between NFT-based experience elements (Murtagh and Contreras, 2012; Griffiths et al., 1984). After applying Ward's method using R, the two-cluster and five-cluster solutions were viable options. However, the five-cluster solution was ultimately selected due to the identified homogeneity of the two-cluster solution revealed through the succeeding observation. We then complemented discriminant analysis to enhance confidence in the result of hierarchical clustering. The result of discriminant analysis is 0.86, which means more than 80 % of the descriptions were correctly classified. In addition, we applied the Fisher ratio to assess class separability and conducted a confidence interval analysis to further examine the robustness of the clustering results (Lin et al., 2004; Shan, 2020). The Fisher ratio was 52.80, indicating strong separation between clusters. The 95 % confidence interval for the classification accuracy was (0.82, 0.84), indicating high reliability and precision in the clustering outcomes.

4. Findings

Our content analysis identified seven purposes of NFT design, five activities using NFTs to interact with consumers and seven reward types that consumers can gain from brand NFTs (see Fig. 1). Our sample's most prominent categories of the brand NFT collections design aim to present brand value (63 %) and ensure brands' presence in the metaverse (56 %). More than half of the purposes of brand NFT design aim to connect with their iconic product (54 %).

The most common activities designed by brands to interact with consumers included event attendance (37 %) and NFT participation (27 %). Brands typically performed combined activities with consumers through NFTs. Our data suggest that 26 % of the brand NFT collections perform multiple activities; 33 % of brand NFT collections show no activities with consumers. Brand NFTs' dominant consumer rewards and gains categories included collectibles (70 %) and physical products (42 %). Consumers gain rewards according to their membership level, which is determined by brand NFT collections.

The distribution of NFT experience adoption varies significantly, with 17 industries identified. Among them, the most obvious trend is that fashion-related industries, including apparel and accessories, footwear, and beauty, account for more than half of the total sample (53 %).

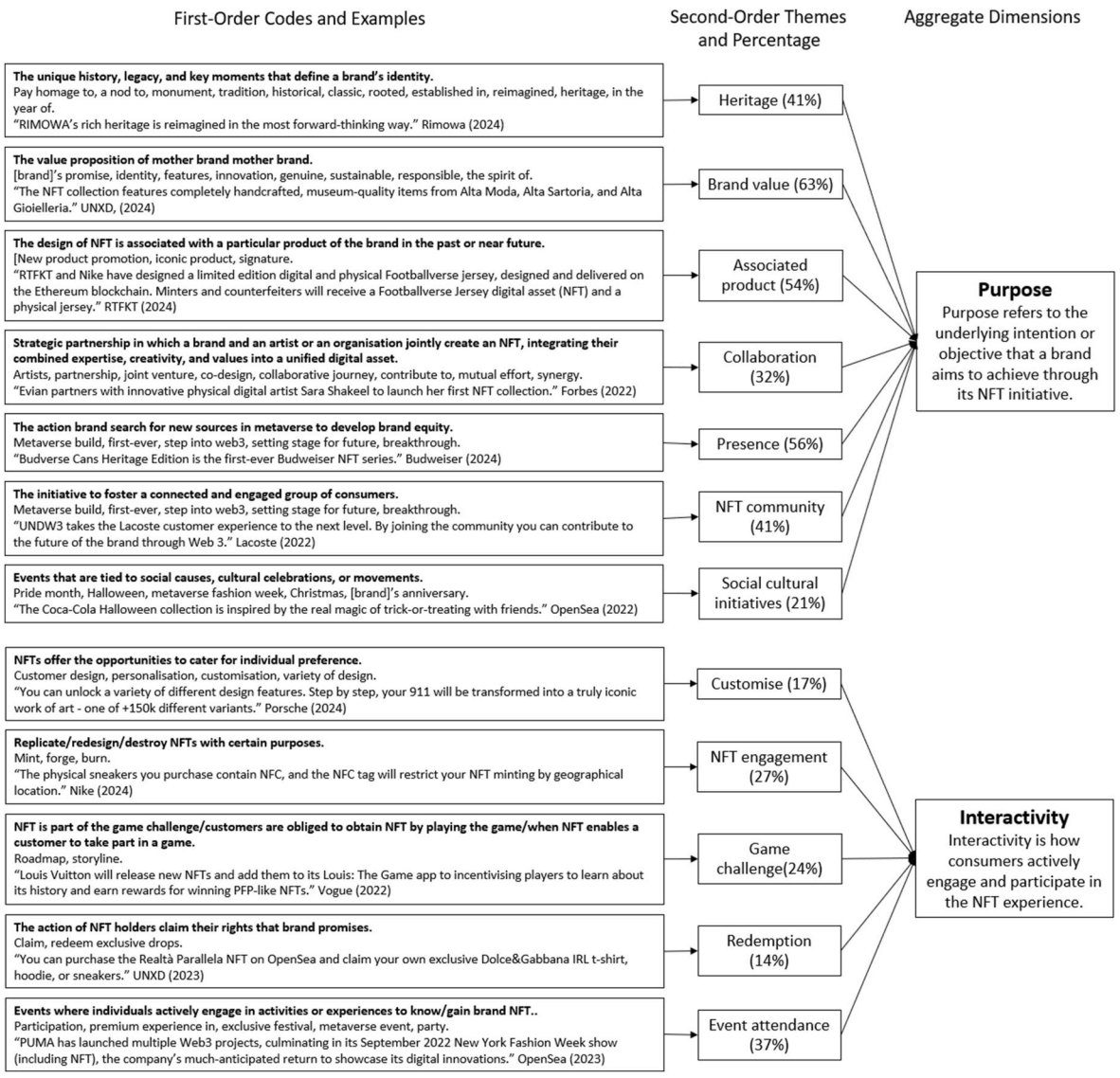


Fig. 1. Content analysis structure

This prominence can be attributed to fashion's natural fit with digital culture, creativity, and exclusivity, enabling brands to leverage NFTs to create visual aesthetics, limited-edition collections, and novel experiences (Zhang and Phang, 2024). Other industries such as food and beverage, automotive, financial services, and publishing have also begun integrating NFTs into their marketing strategies, although to a lesser extent.

Five brand NFT collection types emerged as we conducted cluster analysis on the content analysis results. Table 3 displays the profile of the five types, which are based on purposes, interactivities and reward types.

Our interpretation of the five brand NFT experiences that emerged from the cluster analysis reveals that each strategy uses different branding resources to enable brand marketing in the metaverse. In what follows, we describe the configuration of attributes that brands adopt to develop their digital assets in Web3.

Brand NFTs in the heritage experience cluster are mainly designed to pay homage to their brand history, with NFT launching dates coinciding with special events. Adoption of the heritage type is primarily driven by the food, beverage, and liquor industries (37 %), with NFTs serving as interactive, visual storytelling media of the brand's journey, illustrating important moments, products, or figures that define the brand's identity, bridging consumers' knowledge about the brand's history dynamically and engagingly. Brand history, brand image, expectations and assets constitute the dimensions of brand heritage (Urde et al., 2007). Brand NFTs in this cluster contain at least one of the four dimensions above. Super Puma comics are a good example: as part of its 75th anniversary celebrations, the brand launched a profile NFT collection featuring the iconic character "Super PUMA". The series presents adventures inspired by some of Puma's most memorable sports history over the past 75 years (OpenSea, 2023). By leveraging NFTs, brands embody the characteristics of longevity, continuity, and consistency, essential for maintaining a meaningful brand identity, as highlighted by Cooper et al. (2015). Besides, the NFT with IP rights allows consumers to own a stake in the brand's ecosystem, transforming the feature of consumer engagement with the brand.

Brand NFTs in the community experience cluster focus on brand community building and maintenance in the metaverse. While showing no specific industrial pattern, brands in this cluster primarily use online and offline events to extend brand communities into the virtual world. NFTs act as identity badges in this experience. Not only can they serve as exclusive rewards in virtual spaces for brand-loyal consumers, providing

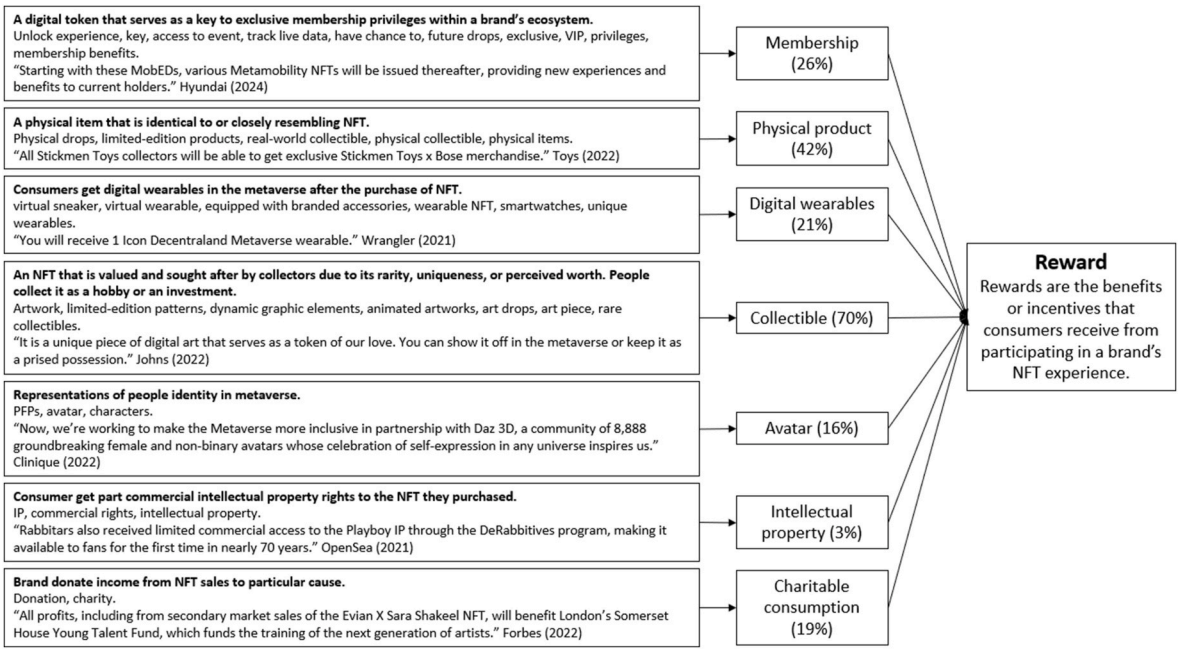


Fig. 1. (*continued*).

Table 3
Clusters of brand NFT experiences.

Clusters	Interpretation	Attributes
Heritage experience	Brands reflect their history and image through NFTs, and the release of NFTs is related to special brand festivals or dates.	Brand value, Brand heritage, Event attendance, Collectible, Intellectual property
Community experience	NFTs build and maintain brand communities in the metaverse, usually rewarding loyal consumers with a physical product.	Presence, NFT community, Membership
Product experience	An NFT released by the brand is related to its existing iconic products or upcoming new products.	Associate product, Special cultural initiatives, NFT engagement, Redemption, Physical product
Collaboration experience	The launch of the brand NFT is usually the brand's first attempt to spread brand value via the metaverse through collaborations with institutions or individual artists.	Collaboration, Charitable consumption
Gamification experience	The acquisition and use of the brand NFT involve some gamification, which is a special form of interaction between brands and consumers	Customise, Game challenge, Digital wearable, Avatar

a sense of recognition and status in digital communities, but through the visual effects of badge markings, NFT enables consumers to demonstrate their connection to the brand both within the digital space and to the wider public. In addition, we found that this type of brand NFT will feed back to consumers with physical products, particularly peripheral products. The DG Family NFT launched by Dolce & Gabbana provides its owner with access to a series of exclusive products, including digital and physical wearable devices (D&GWORLD, 2022). Through the NFT set by member access, Dolce & Gabbana can have real and virtual dialogues with loyal fans and build a multidimensional brand membership community. By leveraging NFTs to build exclusive token communities, brands are able to link access to exclusive content, events, or benefits to ownership of NFTs, representing a unique driver of brand loyalty.

The brand NFT designs in the product-focused experiences are related to the brand's iconic or upcoming products. In this product-focused experience, NFTs serve as product access passes that unlock the ability to purchase or receive physical products, which are mainly adopted by the footwear industry (40 %). Consumers must mint (create) or burn (destroy) an NFT to receive the associated physical product. This creates a unique immersive experience for the brand, where participation in the NFT ecosystem is required to obtain the product. NFT collection of "Space Drip Forging—RTFKT Space Drip x Nike Air Force 1," launched by Nike, is representative of brand NFTs with product strategies (PPTFKT, 2023). The unique engagement mode of NFT allows NFT holders to participate in the co-creation process as well as extra space for personalised product development; it also provides hybrid virtual and physical benefits after the engagement process, such as digital collectibles paired with physical objects.

NFTs in the collaboration experience cluster are launched in collaboration between brands and a third-party, such as artists or institutions. Brands show their ambition to expand their influence in the metaverse through NFTs. NFTs in collaboration experience are usually the brand's first attempt to enter the metaverse by collaborating with parties who have a certain influence in a certain field and donating the final proceeds to specific institutions. Therefore, as a change medallion, an NFT can enable brands to connect with consumers through social responsibility, collaboratively supporting charitable initiatives that enhance their societal impact. By demonstrating a commitment to social causes, NFTs position the brand as a socially conscious entity, aligning its mission with values that resonate with the target audience. Evian partnered with contemporary artist and crystal lover Sara Shakeel to create NFT works and give back to the next generation of future digitalisation, as 100 % of proceeds from the sale of the NFTs will be donated directly to fund training in arts (Forbes, 2022). By working with artists, brands can integrate cultural values into brand NFT. Another type of collaboration is collaboration with other brands or NFT platforms, which is beneficial in expanding the brand's influence by creating mixed communities of both parties and sharing the space for communication through NFT. Besides, collaborations involving NFTs show a linkage to philanthropy, allowing brands to contribute to social welfare directly.

By leveraging NFT, brands are able to effectively raise awareness of social issues by engaging consumers in them.

Finally, brand NFTs represented in the gameplay cluster focus on gamifying NFT. In this cluster, an NFT runs through the brand's game experience. It will be extended to avatars, electronic wearables or brand membership permissions as game rewards to give back to consumers. The apparel and accessories industry brands demonstrate a dominant adoption of this type of NFT integration (68 %). As a gamification element, NFTs are able to integrate gamification features such as rewards and achievements to encourage consumers to interact with the brand in a gameful way.

An example of this is Louis, which is a game software launched by Louis Vuitton in 2021. Louis Vuitton invites consumers to complete challenges in the game to earn their own NFT. Different NFTs can be customised with different Louis Vuitton letter patterns and colours that appear in the game challenges. As a consumer reward, the resulting NFT can be used as a personal avatar in the metaverse (Vogue, 2022). The gamified role of NFT is significant in encouraging positive consumer engagement.

5. Discussion

This exploratory study identifies 19 distinct characteristics and provides an empirical basis for detailing the five NFT-based brand experiences deployed by brands that connect to the developing blockchain market to communicate brand meaning and transform brand experience (brand heritage, community, product, collaboration, and gamification). The five brand experience design aspects reveal the multifaceted nature of brand experiences in the digital era and their alignment with contemporary branding literature.

The introduction of brand NFTs includes a description of the vision for using NFTs, which can be analysed as using NFTs to create a unique set of brand-related associations that brand strategists hope to create or maintain, which can be further transformed into a brand identity. This view is consistent with previous research that indicates that the brand communicates meaning through brand experience (Santos et al., 2022). Our analysis supports this key finding. The adoption of NFT strategies reveals industry-specific patterns, with the most notable trend observed in fashion-related sectors. In addition, the brands integrate the equity potential of NFTs as digital assets into their experiential marketing to facilitate brand experiences on the blockchain. These findings empirically confirm that, regardless of which NFT strategy brands adopt, they effectively and efficiently design brand experiences through NFT activities to address brand identity associations in the blockchain, thereby distinguishing them from other brands. Therefore, based on these findings, this study believes that NFTs play unique roles as the combination of communication tool and assets for brand competitive differentiation and proposes the following conceptualisation. Fig. 2 reveals the effects of NFTs on brand experience by systematically identifying the distinct functions that NFTs serve in the context of brand experience.

In our findings, NFTs are transmitted to consumers as important brand symbols in the experience. An NFT is able to facilitate brand awareness by providing unique, visually engaging digital assets to attract consumers' attention and trigger further interaction. We also found that NFTs are designed by brands to be associated with specific marketing campaigns or product launches; this highlights the brand's involvement in the event and generates excitement around the NFT itself. When these digital assets are shared on social media or other platforms, they can promote the brand and encourage people to discuss its values and products.

Proposition 1. *Brand NFT experience has the potential to enhance brand meaning by improving brand awareness, shaping brand image, and expressing brand personality.*

The importance of multi-channel strategy in digital brand experience is highlighted by the previous research. Building on our findings, NFT plays a key role in facilitating the convergence of brand experience by linking consumers' brand touchpoints. By leveraging NFTs, brands can create hybrid experiences where digital assets enhance physical products or events. This change will redefine how brands cultivate relationships with their audiences, fostering deeper levels of engagement, personalisation, and continuity across multiple dimensions of the consumer journey.

Proposition 2. *Future brand experiences will benefit brand-consumer relationships by bridging the physical and virtual worlds rather than concentrating solely on one.*

As a digital asset, an NFT can be integrated into the brand experience to increase the brand's monetary assets directly. In addition, the support empowered by blockchain enables NFTs to provide brands with a unique way to transfer assets to consumers on the blockchain. Thus, NFTs cultivate a sense of ownership and loyalty by enabling consumers to obtain brand digital assets. In addition, the monetary value of NFTs creates secondary market opportunities, further enhancing the brand's digital ecosystem's long-term financial value and driving continued participation and investment.

Proposition 3. *Integrating NFTs into branding experience could foster long-term value and investment for brands.*

NFTs enable brands to expand their reach beyond traditional consumer groups by appealing to multifaceted audiences. Our research has found that the design of brand NFTs is not limited to the brand's

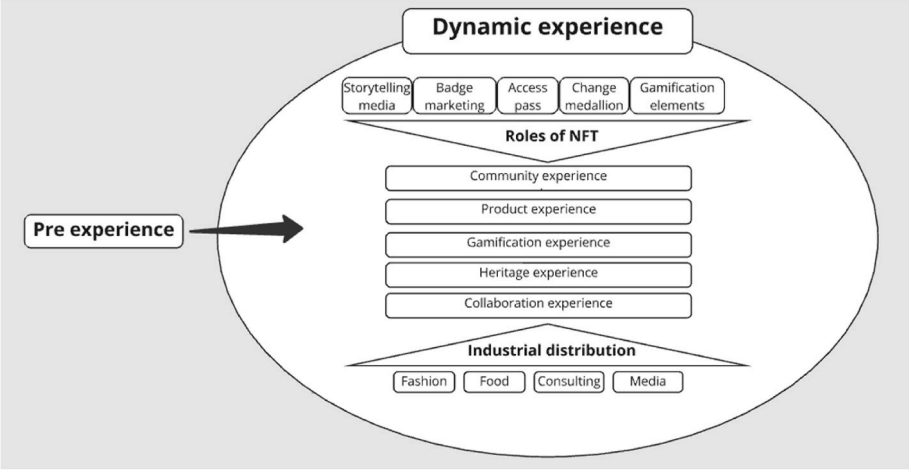


Fig. 2. NFT-based brand experience dynamics

audience. In addition to target audiences, digital collectibles enthusiasts are a potential customer group for the brand. Brands can design NFT experiences to attract those who value exclusivity and rarity, providing them with limited-edition digital assets associated with the brand's value proposition in the virtual world. Gamers represent another important audience group because NFTs can be integrated into the gaming environment to provide in-game assets or rewards to enhance the gaming experience. Bringing different audiences together further enriches the brand's community ecology.

Proposition 4. *Brand NFT experience enables brands to extend their audience beyond traditional consumers, facilitating the reach to diverse groups.*

6. Conclusion

In general, the brand's NFT experience creates avenues for deeper connections with consumers and significantly enhances the potential for value co-creation. This study highlights the dynamic context of the metaverse. The brand NFT is becoming a key tool to encapsulate a brand's history and legacy in an immutable blockchain ledger. As digital assets, these NFTs play roles in brand meaning expression and demonstrate excellent flexibility in integrating brand touchpoints. This flexibility is reflected in the range of reward formats they offer and extends further to include adaptive time applications across the entire consumer journey.

Our investigation reveals that, as a digital asset, an NFT can represent its value and contribute to the presentation and development of brand meaning. Brand adoption of the NFT experience relies on the communication of brand equity between the blockchain and the real world, which is then combined with one or more other platforms. These brand experiences that are communicated and leveraged include brand heritage, community, products, collaborations, and gamification. These findings empirically confirm that whatever NFT experience a brand adopts, its NFT activities can differentiate it from other similar brands in the blockchain, effectively and efficiently develop brand equity and establish comparative advantage. Through NFTs, brands intend to interact with consumers, collectors, and investors, expanding their influence in the digital ecosystem.

6.1. Theoretical implications

Complementing earlier research, our findings first involve the latest momentum in brand practice in blockchain and uncover the specific contributions of NFTs to dynamic experiences in a branding context. We empirically demonstrate how blockchain-based NFTs impact brand experience and identity management in the virtual world. We combined qualitative and quantitative methods to interpret the brand's NFT experience by looking for the brand's judgment on NFTs' value and using it as an anchor point. Our findings provide insights into brand NFT marketing strategies to improve brand experience, which directly responds to the call for in-depth exploration of brand NFT strategy adoption, as highlighted in digital branding literature. Furthermore, our findings contribute to the literature by revealing a trend of brands that use NFTs to combine multiple types of brand experience, leveraging the diverse characteristics of NFTs to create more comprehensive brand experiences in the rapidly evolving virtual space. We also contribute to the literature on consumer behaviour by exploring the diverse dynamics of brand NFT-based experiences.

Second, by examining brand information, our typology fills a gap in the existing literature by providing a classification of brand NFTs. This systematic categorisation could facilitate a better understanding of how to shape consumer experience, offering a structured insight into how NFT can influence brand strategies deployment and consumer behaviour in the context of the metaverse.

Finally, this study contributes by identifying industrial patterns in NFT-based experiences. Illustrations of industry patterns provide insights into how various industries distinctively leverage unique brand opportunities provided by NFTs. Additionally, this study contributes to the digital transformation literature by revealing how NFT influence brand strategies evolution across industries.

6.2. Managerial implications

This study expands our understanding of how brands communicate with consumers, contributing to brand experience management. First, this typology could serve as a guide for brands seeking to design effective NFT experiences. It could also help brands identify which type of NFT experiences align with their overall marketing strategy and brand identity. These types enable brand marketers to shape the utility and value of NFTs based on brand personality and effectively organise and manage their digital assets, including the NFTs themselves. Then, identifying NFT functions in brand experience enables brand managers to decide better how to adopt NFT, enhance their brand's digital presence, and adapt to the ever-changing NFT landscape with agility and precision. NFT's diverse roles in brand strategies provide a framework for brand managers to leverage NFTs effectively to achieve targeted marketing objectives. In addition, brand managers can develop targeted brand strategies by understanding how different industries adopt NFT, enabling brand managers to align the brand messaging with industry-specific NFT practices. This research particularly highlights practical implications for fashion-related industry managers. For example, apparel and accessories show their focus on NFT gamification experiences characterised by game interactions and customisation, while footwear put more attention on associating their offline products to NFTs.

However, the multifaceted role of NFTs in brand experiences could also introduce unanticipated risks to brand-consumer relationships. Their inherent attributes of authenticity, monetary value, and transparency present challenges related to consumer protection, potentially exposing consumers to financial risks and privacy concerns. Therefore, brands should carefully evaluate the experience procedures and formats to responsibly integrate NFTs into their strategies, maximising the benefits of NFTs while minimising risks based on the different types of brand experiences design.

6.3. Limitations and future research

We offer several propositions for future research. More research is needed to understand the impact of NFTs on brand assets, brand identity, and image. For example, future research could explore the comparative effectiveness of brand NFT strategies. Although brand expectations are discussed to some extent in this study, the impact on the target market compared to traditional digital communication, especially on individual consumers and the brand community, has not been tested. Surveys could be designed to reach the target audience group, allowing for the collection of quantitative data on consumer adoption intentions, consumer engagement and demographic influences related to brand NFTs. In the meantime, incorporating in-depth interviews would offer a deeper understanding of the impact of NFTs on brands. This approach could be applied to captures the concerns of consumers and uncover the dynamic of consumer brand attitude caused by NFTs. We also encourage researchers to carefully study the boundary conditions under which NFT functions as a brand's product or channel. For example, we need to understand whether consumers' acceptance of NFTs will affect their perception of the brand and whether adopting this experience generates a positive or negative attitude toward the brand image.

Furthermore, our study only focuses on horizontal classification through existing brands that have already adopted NFTs. Future research is needed to understand whether brands' NFT launches have temporal distribution characteristics from a longitudinal perspective. For example, is the existing NFT experience designed for long-term

interaction or short-term use for special events? Is incorporating an NFT experience becoming pre-release popular among brands?

We use the information released by brands in the NFT market as the main data collection method. On the one hand, it limits our understanding of the multi-channel promotion of NFTs. On the other hand, there is also a lack of data on consumer feedback. This limits our exploration of consumer perceptions. Future classification and evaluation of NFT experiences can refer to these factors and explore the power of media-oriented publicity to study consumers' use of brand NFTs. Another potential issue based on the current data collection method is the risk of incomplete data. While we collect information from verified sources, the rapidly evolving nature of the NFT market suggests that brand NFT initiatives launched after our search data may not have been captured. A longitudinal study could offer significant value, providing a more dynamic perspective and validating the proposed typology as the NFT landscape continues to evolve.

Our study's focus on brand NFT design excludes the consideration of external factors, such as evolving NFT regulations, consumer sentiment, which may impact the findings' relevance. To address this limitation, future research could incorporate expert interviews to explore how external factors shape the applicability of NFT-based strategies.

Finally, the generalisability of our findings is limited. Our analysis helps identify certain characteristics and functions of brand NFTs from an experiential development perspective. Future research is needed to explore how and under what conditions different experiences can be adopted efficiently. This should consider the brand's resources and the broader ecosystem within which they adopt the experience mode. This research lays a foundation for further investigating the brand management mechanism using blockchain technology through brand NFTs.

CRediT authorship contribution statement

Wenjie Li: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Graciela Corral de Zubielqui:** Writing – review & editing, Validation, Supervision, Project administration. **Sally Rao Hill:** Writing – review & editing, Validation, Supervision.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

We have nothing to declare.

Acknowledgement

The authors would like to express our sincere gratitude to all those who have contributed to the successful completion of this research paper.

We are grateful to our colleagues and peers who have provided valuable discussions and feedback during various stages of this research. Thank you for being an integral part of this journey.

Appendix A. NFT collections

No	Brand	Industry	collections	Source
1	Nike	Footwear	RTFKT X NIKE Trillium Lace Engine	marketplace: Opensea, news websites: CoinStats collaborative platform: RTFKT website
2			RTFKT x Nike Dunk Genesis CRYPTOKICKS	marketplace: Opensea, news websites: CoinGecko collaborative platform: RTFKT website
3			RTFKT x NIKE AR HOODIE PRE FORGED	marketplace: Opensea, social media: Instagram collaborative platform: RTFKT website
4			RTFKT x Nike Footballverse Jersey	marketplace: Opensea, Rarible, collaborative platform: RTFKT website
5			Space Drip Forging - RTFKT Space Drip x Nike Air Force 1	marketplace: Opensea, NFTGo collaborative platform: RTFKT website
6	Dolce & Gabbana	Apparel and Accessories	Dolce&Gabbana: DGFamily Glass Box	marketplace: Opensea, NFTGo collaborative platform: UNXD website
7			Dolce&Gabbana: DGFamily	marketplace: Opensea, news websites: CoinDesk, collaborative platform: UNXD website
8			inBetweeners x Dolce&Gabbana Drip	marketplace: Opensea, news websites: CoinStats, social media: X
9			Dolce&Gabbana Disco Drip	marketplace: Opensea, news websites: NFTstats, NFT Plazas
10			Dolce&Gabbana Realta Parallela	marketplace: Opensea, news websites: Coinbase NFT, collaborative platform: UNXD website
11			Collezione Genesi	Marketplace: Opensea, social media: Twitter, news website: NFT now
12	Tiffany	Watch and jewellery	NFTiff	marketplace: Opensea, news websites: NFT evening, social media: X
13	Gucci	Apparel and Accessories	The Next 100 Years of Gucci	Brand website, news websites: CoinStats collaborative platform: RTFKT website
14			10KTF Gucci Grail mint pass	marketplace: Opensea, brand website, social media: X
15			10KTF Gucci Grail	marketplace: Opensea, brand website, social media: X
16			supergucci	Brand website, marketplace: Opensea, news website: NFT Canlendar
17	Adidas	Footwear	adidas Originals Into the Metaverse	marketplace: Opensea, brand website, social media: X
18			adidas for Prada resource	marketplace: Opensea, brand website, news website: NFT now
19			adidas Virtual Gear - Genesis Collection	marketplace: Opensea, brand website, news website: CoinStats
20			Raws x adidas | Through the Stripes	marketplace: Opensea, Rarible, news website: NFT Valuations
21	McLaren	Automobile	McLaren MSO LAB Genesis	marketplace: Opensea, Rarible, brand website
22	Pepsi	Food and beverage	Pepsi Mic Drop	marketplace: Opensea, Rarible, social media: X
23	Budweiser	Food and beverage	Budverse Cans - Heritage Edition	marketplace: Opensea, Rarible, brand website
24			Budverse Live Scoreboard Collectible 2022	marketplace: Opensea, Rarible, news website: NFT Evening

(*continued on next page*)

(*continued*)

No	Brand	Industry	collections	Source
25			Budverse Legends	marketplace: Opensea, brand website, news website: NFT calendar
26			Zed Run - Budweiser Pass	marketplace: Opensea, brand website, collaborative platform: zedrun community
27	Lacoste	Apparel and Accessories	Lacoste UNDW3:The Emerge (round2)	marketplace: Opensea, Rarible, brand website
28			Lacoste UNDW3:Genesis Pass (round1)	marketplace: Opensea, news website: NFT now, brand website
29	Coca cola	Food and beverage	The Coca-Cola Friendship Day Collection	marketplace: Opensea, brand website, news website: NFT Evening
30			Coca-Cola Pride Collection	marketplace: Opensea, brand website, news website: NFT Evening
31			The Coca-Cola Halloween Collection	marketplace: Opensea, brand website, news website: cryptonews
32	Nivea	beauty supplies	the value of touch	marketplace: Opensea, Social media: YouTube, brand website
33	Prada	Apparel and Accessories	Prada Timecapsule	marketplace: Opensea, news website: Forbes, brand website
34	Stella Artois	Food and beverage	Stella Artois - Racing in THE LIFE ARTOIS	marketplace: Opensea, news website: Forbes, NFT calendar
35	Puma	Footwear	Super puma	marketplace: Opensea, news website: Forbes, brand website
36			Super puma comics	marketplace: Opensea, news website: Forbes, NFT Plazas
37			Nitro Collection by PUMA	marketplace: Opensea, Nifty gateway studio, brand website
38			Nitro Token by PUMA	marketplace: Opensea, Nifty gateway studio, social media: YouTube
39	Clinique	beauty supplies	Clinique x Daz 3D - Metaverse More Like Us	marketplace: Opensea, news website: Forbes, brand website
40	Audi	Automobile	Unseen Universe Official	marketplace: Opensea, DappRadar, Social media: Facebook
41	Playboy	Periodical Publisher	Playboy rabbitars official	marketplace: Opensea, DappRadar, Social media: Instagram
42			PLAYBOY x Slimesunday Liquid Summer	marketplace: Opensea, NFTGo, Social media: YouTube
43			The Playboy Party People	Blockchain platform: Sandbox, marketplace: Opensea, DappRadar
44	Starbucks	Food and beverage	Starbucks the first store collection	marketplace: Opensea, Nifty gateway studio, news website: CoinDesk
45			The Starbucks Siren Collection	marketplace: Opensea, Nifty gateway studio, news website: CoinDesk
46	Vogue	Periodical Publisher	Vogue Singapore NFT Collection	marketplace: Opensea, Rarible, brand website
47			Every Body by Vogue Singapore x The MetaArt Club	marketplace: Opensea, news website: NFT plazas, brand website
48			Fashion for Peace NFT Collection by Vogue Singapore x Vogue Ukraine	marketplace: Opensea, news website: NFT calendar, brand website
49	Time	Periodical Publisher	TIMEPieces Build a Better Future: Genesis Drop	marketplace: Opensea, NFT valuation, news website: Time magazine
50	Renault	Automobile	R3NLT-genR5	marketplace: Opensea, social media: X, brand website
51	Hugo	Apparel and Accessories	HUGO x Imaginary Ones: Embrace Your Emotions	marketplace: opensea, news website: NFT calendar, brand website
52	Hyundai	Automobile	Hyundai Metamobility - MobED	marketplace: Opensea, NFTGo, Magic Eden
53			Hyundai Metamobility - IONIQ 6	marketplace: Opensea, NFTGo, brand website
54			Hyundai X Meta Kongz	marketplace: Opensea, brand website, social media: YouTube
55	Diesel	Apparel and Accessories	Diesel x Hape: Antidote to the Status Quo	marketplace: Opensea, DappRadar, social media: X
56	Club soda	Food and beverage	Club Soda 3.0 Founder's Pass	marketplace: Opensea, CoinGecko, social media: LinkedIn
57	Mac	beauty supplies	MAC VivaGlam	brand website, news website: Vogue business, marketplace: Opensea
58	Balmain	Apparel and Accessories	Balmain x Space Runners Unicorn	brand website, news website: Vogue business, marketplace: Rarible
59			The Balmain Thread	news website: NFT lately, NFT news today, brand website
60			Balmain x Barbie	brand website, market place: mattelcreations, news website: Forbes
61	Bose	Electronic	Stickmen Toys	marketplace: Opensea, social media: X, news website: NFT evening
62	Australian Open	Sports event services	AO ArtBall	marketplace: Opensea, social media: X, brand website
63	Reddit	Internet Publishing	Reddit Collectible Avatars	marketplace: Opensea, brand website, news website: NFT evening
64	Louis Moinet	Watch and jewellery	Louis Moinet Space Revolution	marketplace: Opensea, DappRadar, Coinstats
65	Timex	Watch and jewellery	Timex x Bored Ape Community Forge Pass	marketplace: Opensea, brand website, news website: NFT evening
66	airBaltic	Transportation	airBaltic City Collection	marketplace: Opensea, brand website, social media: Facebook
67	Rimowa	Apparel and Accessories	RTFKT x RIMOWA Meta-Artisan Collection	marketplace: Opensea, Rarible, coinstats
68	Louis Vuitton	Apparel and Accessories	Louis: the game	marketplace: Opensea, news website: blockworks, NFT evening
69	Taco bell	Food and beverage	Taco bell	marketplace: Rarible, Opensea, news website: business insider
70	Pizza hut	Food and beverage	Pizza hut	marketplace: Rarible, news website: trend hunter, pr newswire
71	Burger King	Food and beverage	keep it real meats (burger king)	brand website, social media: LinkedIn, marketplace: Opensea
72	Warner	Picture and Video production	Warner	news website: NFT calendar, NFT evening, social media: X
73	Lamborghini	Automobile	Lamborghini	brand website, marketplace: CoinDesk, social media: Instagram
74	FIFA	Sports event services	fifa collect	brand website, blockchain platform: upland, news website: NFT evening
75	McDonald	Food and beverage	McRib (Mcdonald)	marketplace: Opensea, news website: NBC news, NFT evening
76	Evian	Food and beverage	evian	brand website, news website: Forbes, social media: LinkedIn
77	Gap	Apparel and Accessories	gap	brand website, news website: the verge, NFT calendar
78	Royal Salute	Food and beverage	royal salute	brand website, news website: The Whiskey Wash, marketplace: blockbar
79	Papa John's	Food and beverage	papa john's	brand website, news website: NFT lately, NFT culture
80	Frank's Redhot	Food and beverage	redhot	marketplace: Opensea, social media: X, Instagram
81	Foreo	Beauty supplies	foreo	brand website, news website: WWD, social media, Facebook
82	Redbull	Food and beverage	redbull	Brand website, news website, sporting crypto, the block
83	UFC	Sports event services	UFC	brand website, blockchain platform: crypto, social media: reddit
84	NFL	Sports event services	NFL	social media: reddit, news website: NFT plazas, blockchain news

(*continued on next page*)

(*continued*)

No	Brand	Industry	collections	Source
85	Aston Martin	Automobile	Aston Martin	brand website, blockchain platform: crypto, marketplace: magic eden
86	Acura	Automobile	acura	brand website, news website: marketing dive, motor trend
87	IWC	Watch and jewellery	iwc	brand website, blockchain platform: Arianee, news website: Luxury Tribune
88	Afterpay	Financial service	afterpay	news website: retail dive, Hypebeast, ledger insights
89	Wrangler	Apparel and Accessories	wrangler	brand website, news website: business wire, NFT evening
90	Burberry	Apparel and Accessories	Sharky B	social media: Facebook, marketplace: CoinStats, news website: Mythical News
91			Minny B	brand website, news website: Forbes, news website: NFT calendar
92	Porsche	Automobile	PORSCHΞ 911	Marketplace: Opensea, brand website, social media: Twitter
93	KARL LAGERFELD	Apparel and Accessories	KL7 X ENDLESS	brand website, news website: milled, fashion website: wwd
94	Jimmy Choo	Footwear	Eric Haze Chasing Stars	brand website, news website: sohu, Web3 platform
95			Jimmy Choo Mystery Box	brand website, news website: sohu, Web3 platform
96	Bulgari	Watch and jewellery	Serpenti Metamorphosis	social media: Facebook, brand website, social media: X
97	Ray-Ban	Eyewear	RAY-BAN x EXTRAWEG	news website: NFT stats, marketplace: Opensea, social media: X, Facebook
98	Givenchy	Apparel and Accessories	CHITO X GIVENCHY NFT	social media: X, marketplace: Opensea, brand website
99	Coach	Apparel and Accessories	Coach Holiday Animals	social media: twitter, fashion website: wwd, news website: trendhunter
100	Patrón	Food and beverage	PATRÓN en LALIQUE: Serie 3	news website: Forbes, marketplace: blockbar, news website: spirits business
101	Jacob & Co.	Watch and jewellery	Astronomia Metaverso Collection	news website: Forbes, marketplace: sbnift market, brand website
102	Hublot	Watch and jewellery	Hublot x Takashi Murakami NFTs	Brand website, fashion website: highsnobiety, socail media: TikTok
103	Salvatore Ferragamo	Apparel and Accessories	NFT Installation	fashion website: vogue, news website: Forbes, fashion website: v magazine
104	Tag Heuer	Watch and jewellery	Calibre E4 smartwatch	news website: NFTandGameFi, brand website, news website: Forbes
105	Balenciaga	Apparel and Accessories	To the Moon NFT	news website: NFT calendar, NFT evening, market place: crypto
106	Orientalist Spirits	Food and beverage	The Founders Collective	social media: X, spirit website: the spirit business,
107	Dewar's	Food and beverage	Dewar's Double Double 21-Year-Old Rye Cask Finish	marketplace: blockbar, Opensea, industry website: the whiskey wash,
108	Hennessy	Food and beverage	Cafe 11 NFT	marketplace: blockbar, Opensea, brand website
109	Johnnie Walker	Food and beverage	Johnnie Walker Masters of Flavour	marketplace: blockbar, industrial website: the spirit business, news website: adnews
110	Yves Saint Laurent	beauty supplies	YSL Beauty Golden Blocks	brand website, marketplace: Opensea, blockchain platform: arianee
111	L'Oréal Paris	beauty supplies	The Reds of Worth NFT	brand website, marketplace: Opensea, news website: coindesk
112	NARS	beauty supplies	ORGASM, ACTIVATED	social media: Instagram, news website: nftevening, nftnow
113	Pringles	Food and beverage	CryptoCrisp	marketplace: Rarible, Opensea, social media: X
114	BMW	Automobile	Museum of Sounds	marketplace: Opensea, news website: nfteveing, nft culture
115	Mercedes-Benz	Automobile	The Era of Technology	social media: X, news website: nft calendar, nft evening
116	Under amour	Apparel and Accessories	Genesis Curry Flow NFT	marketplace: Opensea, news website: coindesk,
117	Levis	Apparel and Accessories	Levi's® 501®	marketplace: opensea, social media: Facebook, brand website
118	Disney	Picture and Video production	Golden moments	marketplace: veve, social media: raddit, news website: nft now

Data availability

The data samples used in this study are provided in the appendix for transparency.

References

Adegbola, O., Gearhart, S., Skarda-Mitchell, J., 2018. Using Instagram to Engage with (Potential) Consumers: A Study of Forbes Most Valuable Brands' Use of Instagram, vol. 7. The Journal of Social Media in Society, pp. 232–251.

Aldenderfer, M., Blashfield, R., 1984. Cluster Analysis. A SAGE University Paper, Newbury Park.

Alexa, M., 1997. Computer-assisted Text Analysis Methodology in the Social Sciences.

Alexander, B., Bellandi, N., 2022. Limited or limitless? Exploring the potential of NFTs on value creation in luxury fashion. Fash. Pract. 14, 376–400.

Ante, L., 2023. Non-fungible token (NFT) markets on the Ethereum blockchain: temporal development, cointegration and interrelations. Econ. Innovat. N. Technol. 32, 1216–1234.

Bailey, K.D., 1994. Typologies and Taxonomies: an Introduction to Classification Techniques. Sage.

Bao, W., Hudders, L., Yu, S., Beuckels, E., 2024. Virtual luxury in the metaverse: NFT-enabled value recreation in luxury brands. Int. J. Res. Market.

Barrera, K.G., Shah, D., 2023. Marketing in the Metaverse: conceptual understanding, framework, and research agenda. J. Bus. Res. 155, 113420.

Bilgihan, A., Leong, A.M.W., Okumus, F., Bai, J., 2024. Proposing a metaverse engagement model for brand development. J. Retailing Consum. Serv. 78, 103781.

Boukis, A., 2024. The effect of tokenized rewards on customer loyalty programs. Ann. Tourism Res. 109, 103851.

Brakus, J.J., Schmitt, B.H., Zarantonello, L., 2009. Brand experience: what is it? How is it measured? Does it affect loyalty? J. Market. 73, 52–68.

Brouard, M., 2024. Setting sail in a tokenized world: an exploration of the Bored Ape Yacht club and the co-created consumer experience. Int. J. Res. Market.

Burla, L., Knierim, B., Barth, J., Liewald, K., Duetz, M., Abel, T., 2008. From text to codings: intercoder reliability assessment in qualitative content analysis. Nurs. Res. 57, 113–117.

Cappellini, B., Yen, D.A.-W., 2016. A space of one's own: spatial and identity liminality in an online community of mothers. J. Market. Manag. 32, 1260–1283.

Chalmers, D., Fisch, C., Matthews, R., Quinn, W., Recker, J., 2022. Beyond the bubble: will NFTs and digital proof of ownership empower creative industry entrepreneurs? J. Bus. Ventur. Insights 17, e00309.

Chandra, Y., 2022. Non-fungible token-enabled entrepreneurship: a conceptual framework. J. Bus. Ventur. Insights 18, e00323.

Chen, Q., Choi, B.-J., Lee, S.-J., 2025. Tailoring customer segmentation strategies for luxury brands in the NFT market–The case of SUPERGUCCI. J. Retailing Consum. Serv. 82, 104121.

Chen, Y., Ye, Y., Zeng, W., 2023. The rich, the poor, and the ugly: an aesthetic-perspective assessment of NFT values. Proceedings of the 16th International Symposium on Visual Information Communication and Interaction, pp. 1–8.

Chicotsky, B., 2023. Web3 and marketing: the new frontier. Appl. Market. Analytic. 9, 182–194.

Cho, M., Ko, E., Taylor, C.R., 2024a. Do non-fungible tokens create long-term value for luxury brands? The effect of NFT promotions on customer equity. Comput. Hum. Behav., 108347
Cho, M., Ko, E., Taylor, C.R., 2024b. The use of non-fungible tokens (NFTs) as promotional devices to build brands: an application of lovemark theory. J. Glob. Fashion Market. 1–17.
Chohan, R., Paschen, J., 2023. NFT marketing: how marketers can use nonfungible tokens in their campaigns. Bus. Horiz. 66, 43–50.
Chohan, U.W., 2021. Non-fungible Tokens: Blockchains, Scarcity, and Value. *Non-fungible Tokens*. Routledge.
Cohen, S., Fishbain, B., 2025. Advancing Metaverse's experience through optimization of players' decisions. Int. J. Inform. Manag. .Data Insights 5, 100316.
Colicev, A., 2023. How can non-fungible tokens bring value to brands. Int. J. Res. Market. 40, 30–37.
Cooper, H., Merrilees, B., Miller, D., 2015. Corporate heritage brand management: corporate heritage brands versus contemporary corporate brands. J. Brand Manag. 22, 412–430.
D&GWORLD, 2022. #DGFamily NFT community - world - Dolce & Gabbana [Online]. Available: https://world.dolcegabbana.com/news/dgfamily-nft-community [Accessed].
Darveau, J., Cheikh-Ammar, M., 2021. The interplay between liminality and consumption: a systematic literature review with a future research agenda. Int. J. Consum. Stud. 45, 867–888.
Das, D., Bose, P., Ruaro, N., Kruegel, C., Vigna, G., 2022. Understanding security issues in the NFT ecosystem. Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security, pp. 667–681.
Daskalaki, M., Simosi, M., 2018. Unemployment as a liminoid phenomenon: identity trajectories in times of crisis. Hum. Relat. 71, 1153–1178.
Davis, F.D., 1985. A Technology Acceptance Model for Empirically Testing New End-User Information Systems: Theory and Results. Massachusetts Institute of Technology.
Dwivedi, Y.K., Hughes, L., Baabdullah, A.M., Ribeiro-Navarrete, S., Giannakis, M., Al-Debei, M.M., Dennehy, D., Metri, B., Buhalis, D., Cheung, C.M., 2022. Metaverse beyond the hype: multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. Int. J. Inf. Manag. 66, 102542.
Eastman, J., Yazdanparast, A., Ketron, S., 2024. Identifying NFT consumer segments: a consumption value theory and luxury perspective. Market. Intell. Plann. 42, 1581–1600.
Elo, S., Kyngäs, H., 2008. The qualitative content analysis process. J. Adv. Nurs. 62, 107–115.
Fiss, P. C., 2007. A set-theoretic approach to organizational configurations. Acad. Manag. Rev. 32, 1180–1198.
Florès, L., Lejealle, C., 2023. Web 3 and NFT create value to the wine industry: the case of Chateau Edmus. J. Inf. Technol. Teach. Cases, 20438869231219517.
Flynn, B.B., Huo, B., Zhao, X., 2010. The impact of supply chain integration on performance: a contingency and configuration approach. J. Oper. Manag. 28, 58–71.
FORBES, 2022. Fashion's favorite water partner mints charity NFTs with buzzy phygital artist Sara Shakeel [Online]. Available: https://www.forbes.com/sites/stephaniehirschmiller/2022/04/04/evian-x-sara-shakeel-nfts-minted-on-tezos-blockchain/ [Accessed].
FORBES, 2024. Forbes 2024 global 2000. List [Online]. Available: https://www.forbes.com/lists/global2000/.
Gagné, M., Deci, E.L., 2005. Self-determination theory and work motivation. J. Organ. Behav. 26, 331–362.
Getchell, K.M., Beitelspacher, L.S., 2020. Better marketing for female marketers: gendered language in the Forbes CMO list. Bus. Horiz. 63, 607–617.
Griffiths, A., Robinson, L.A., Willett, P., 1984. Hierarchic agglomerative clustering methods for automatic document classification. J. Doc. 40, 175–205.
Gupta, Y., Kumar, J., Reifers, A., 2022. Identifying security risks in NFT platforms. arXiv preprint arXiv:2204.01487.
Hackley, C., Hackley, R.A., Bassiouni, D.H., 2021. Imaginary futures: liminoid advertising and consumer identity. J. Market. Commun. 27, 269–283.
Hakkarainen, T., Colicev, A., 2023. Blockchain-enabled advances (BEAs): implications for consumers and brands. J. Bus. Res. 160, 113763.
Harms, R., Kraus, S., Schwarz, E., 2009. The suitability of the configuration approach in entrepreneurship research. Enterpren. Reg. Dev. 21, 25–49.
Hayes, A.F., Krippendorff, K., 2007. Answering the call for a standard reliability measure for coding data. Commun. Methods Meas. 1, 77–89.
Henry, D., Dymnicki, A.B., Mohatt, N., Allen, J., Kelly, J.G., 2015. Clustering methods with qualitative data: a mixed-methods approach for prevention research with small samples. Prev. Sci. 16, 1007–1016.
Hepola, J., Karjaluoto, H., Hintikka, A., 2017. The effect of sensory brand experience and involvement on brand equity directly and indirectly through consumer brand engagement. J. Prod. Brand Manag. 26, 282–293.
Hofstetter, R., De Bellis, E., Brandes, L., Clegg, M., Lamberton, C., Reibstein, D., Rohlfsen, F., Schmitt, B., Zhang, J.Z., 2022. Crypto-marketing: how non-fungible tokens (NFTs) challenge traditional marketing. Mark. (Lond.) 33, 705–711.
Joy, A., Zhu, Y., Peña, C., Brouard, M., 2022. Digital future of luxury brands: metaverse, digital fashion, and non-fungible tokens. Strateg. Change 31, 337–343.
Jung, Y., Pawlowski, S.D., 2014. Virtual goods, real goals: exploring means-end goal structures of consumers in social virtual worlds. Inf. Manag. 51, 520–531.
Kau, A.K., Tang, Y.E., Ghose, S., 2003. Typology of online shoppers. J. Consum. Market. 20, 139–156.
Kaur, J., Mogaji, E., Paliwal, M., Jha, S., Agarwal, S., Mogaji, S.A., 2024. Consumer behavior in the metaverse. J. Consum. Behav. 23, 1720–1738.
Kim, J., Cho, A., Lee, D.C., Park, J., Kim, A., Jhang, J., Kim, C., 2025. Consumer preferences for the visual presentation of non-fungible tokens (NFTs) of luxury products: the role of perceived authenticity. J. Retailing Consum. Serv. 82, 104131.
Kshetri, N., 2022. Securing Trust and Exclusivity: the Role of Blockchain in Luxury Brand Strategies. Available at: SSRN 5005946.
Kumar, A., Shankar, A., Behl, A., Wamba, S.F., 2025. Do you believe in the metaverse NFTs? Understanding the value proposition of NFTs in the metaverse. Technol. Forecast. Soc. Change 210, 123880.
Laver, C., Mcgrath, L., Liebert, R.J., Noorani, T., Barnes, N., Chase, M., Hall, J., Wakeling, B., 2022. 'You don't take things too seriously or un-seriously': beyond recovery to liminal and liminoid possibility in a community arts and mental health project. J. Community Appl. Soc. Psychol. 32, 653–664.
Lee, C.T., Ho, T.-Y., Xie, H.-H., 2023. Building brand engagement in metaverse commerce: the role of branded non-fungible tokens (BNFTs). Electron. Commer. Res. Appl. 58, 101248.
Lee, C.T., Shen, Y.-C., Li, Z., Xie, H.-H., 2024a. The effects of non-fungible token platform affordances on customer loyalty: a Buyer–Creator duality perspective. Comput. Hum. Behav. 151, 108013.
Lee, H., Rana, M.R.I., Xu, Y., 2024b. Transforming luxury: young consumers' motivations towards purchasing virtual luxury non-fungible token wearables. Int. J. Retail Distrib. Manag.
Leigh, T.W., Peters, C., Shelton, J., 2006. The consumer quest for authenticity: the multiplicity of meanings within the MG subculture of consumption. J. Acad. Market. Sci. 34, 481–493.
Lin, T.-H., Li, H.-T., Tsai, K.-C., 2004. Implementing the Fisher's discriminant ratio in ak-means clustering algorithm for feature selection and data set trimming. J. Chem. Inf. Comput. Sci. 44, 76–87.
Mair, J., Battilana, J., Cardenas, J., 2012. Organizing for society: a typology of social entrepreneuring models. J. Bus. Ethics 111, 353–373.
Mao, J., Xu, X., Han, J., Ko, E., 2024. The impact of digital fashion marketing on purchase intention. Asia Pac. J. Mark. Logist.
Massi, M., Piancatelli, C., Vocino, A., 2023. Authentic omnichannel: providing consumers with a seamless brand experience through authenticity. Psychol. Market. 40, 1280–1298.
Mckercher, B., Lui, S.L., 2014. Becoming a tourist. Int. J. Tourism Res. 16, 399–406.
Mekacher, A., Bracci, A., Nadini, M., Martino, M., Alessandretti, L., Aiello, L.M., Baronchelli, A., 2022. Heterogeneous rarity patterns drive price dynamics in NFT collections. Sci. Rep. 12, 13890.
Miller, D., 1996. Configurations revisited. Strateg. Manag. J. 17, 505–512.
Mishra, M., Paul, J., Czinkota, M., 2022. Revisiting models of internationalization: pre-export phase and lateral rigidity of emerging market Small and Medium Enterprises. Thunderbird Int. Bus. Rev. 64, 125–138.
Morgan, D.L., 1993. Qualitative content analysis: a guide to paths not taken. Qual. Health Res. 3, 112–121.
Mostafa, R.B., Kasamani, T., 2021. Brand experience and brand loyalty: is it a matter of emotions? Asia Pac. J. Mark. Logist. 33, 1033–1051.
Murtagh, F., Contreras, P., 2012. Algorithms for hierarchical clustering: an overview. Wiley Interdisciplin. Rev.: Data Min. Knowl. Discov. 2, 86–97.
Murtas, G., Pedeliento, G., Mangiò, F., 2024. Luxury fashion brands at the gates of the Web 3.0: an analysis of early experimentations with NFTs and the metaverse. J. Glob. Fashion Market. 15, 90–114.
O'Connor, C., Joffe, H., 2020. Intercoder reliability in qualitative research: debates and practical guidelines. Int. J. Qual. Methods 19, 1609406919899220.
Ong, C.H., Lee, H.W., Ramayah, T., 2018. Impact of brand experience on loyalty. J. Hospit. Market. Manag. 27, 755–774.
OPENSEA, 2023. Super PUMA comics [Online]. Available: https://opensea.io/collection/super-puma-comics [Accessed].
Ortiz-Avram, D., Ovcharova, N., Engelmann, A., 2024. Dynamic capabilities for sustainability: toward a typology based on dimensions of sustainability-oriented innovation and stakeholder integration. Bus. Strat. Environ. 33, 2969–3004.
Paik, H., Lee, J.H., 2021. Analytical framework, typology and retail experience design process for integrated relational brand experience. Int. J. Retail Distrib. Manag. 49, 466–490.
Park, H., Lim, R.E., 2023. Fashion and the metaverse: clarifying the domain and establishing a research agenda. J. Retailing Consum. Serv. 74, 103413.
Peres, R., Schreier, M., Schweidel, D.A., Sorescu, A., 2023. Blockchain Meets Marketing: Opportunities, Threats, and Avenues for Future Research. Elsevier.
Piasecki, J., Waligora, M., Dranseika, V., 2018. Google search as an additional source in systematic reviews. Sci. Eng. Ethics 24, 809–810.
PPTFKT, 2023. Rtfkt [Online]. Available: https://rtfkt.com/product/63EA7D5AE98363FD50CCC27F?from=tagId.
Prasad, V.V., Murthy, M.D.P., Joseph, J., Adhikari, A., 2023. What decides my purchase of non-fungible tokens? J. Consum. Behav.
Prentice, C., Wang, X., Loureiro, S.M.C., 2019. The influence of brand experience and service quality on customer engagement. J. Retailing Consum. Serv. 50, 50–59.
Ramadan, Z., 2024. The rise of the egocentric brand and consumer: the case of soulbound tokens (SBT). Qual. Mark. Res. Int. J. 27, 921–940.
Rejeb, A., Keogh, J.G., Treiblmaier, H., 2020. How blockchain technology can benefit marketing: six pending research areas. Front. Blockchain 3, 3.
Reshef Kera, D., Eichstetter, E., Ellul, J., 2023. Volumetric and edible NFTs: explorations of the limits of aesthetic assetization and tokenization. Proceedings of the 11th International Conference on Digital and Interactive Arts, pp. 1–6.
Rodríguez-Torrico, P., Trabold Apadula, L., San-Martín, S., San José Cabezudo, R., 2020. Have an omnichannel seamless interaction experience! Dimensions and effect on consumer satisfaction. J. Market. Manag. 36, 1731–1761.

Santos, Z.R., Cheung, C.M., Coelho, P.S., Rita, P., 2022. Consumer engagement in social media brand communities: a literature review. Int. J. Inf. Manag. 63, 102457.
Sarason, Y., Tegarden, L.F., 2003. The erosion of the competitive advantage of strategic planning: a configuration theory and resource based view. J. Bus. Manag. 9, 1–20.
Scott, R., Cayla, J., Cova, B., 2017. Selling pain to the saturated self. J. Consum. Res. 44, 22–43.
Shan, G., 2020. Accurate confidence intervals for proportion in studies with clustered binary outcome. Stat. Methods Med. Res. 29, 3006–3018.
Snelson, C.L., 2016. Qualitative and mixed methods social media research: a review of the literature. Int. J. Qual. Methods 15, 1609406915624574.
SPRI, 2021. Metaverse begins: go to global [Online]. Available: https://spri.kr/spri/eng_index.html [Accessed].
Stansfield, C., Dickson, K., Bangpan, M., 2016. Exploring issues in the conduct of website searching and other online sources for systematic reviews: how can we be systematic? Syst. Rev. 5, 1–9.
Taheri, B., Farrington, T., Gori, K., Hogg, G., O'Gorman, K.D., 2017. Escape, entitlement, and experience: liminoid motivators within commercial hospitality. Int. J. Contemp. Hospit. Manag. 29, 1148–1166.
Taheri, B., Gori, K., O'Gorman, K., Hogg, G., Farrington, T., 2016. Experiential liminoid consumption: the case of nightclubbing. J. Market. Manag. 32, 19–43.
Tan, Y., 2024. 'Capturing the shining moments': performing liminoid Seichi Junrei on social media. Curr. Issues Tourism 27, 1959–1972.
Turner, V., 1983. Liminal to liminoid, in play, flow, and ritual. An Essay in Comparative Symbology.
Turner, V., 1998. From Ritual to Theatre: the Human Seriousness of Play. Performing arts journal publ.
Van der Westhuizen, L.-M., 2018. Brand loyalty: exploring self-brand connection and brand experience. J. Prod. Brand Manag. 27, 172–184.
View, G., 2024. Non-fungible token market size, share & trends analysis report by application (art, sports), by type (physical assets, digital assets), by end-use (commercial, personal), by region, and segment forecasts, 2024 - 2030 [Online]. Available: https://www.grandviewresearch.com/industry-analysis/non-fungible-token-market-report [Accessed].
VOGUE, 2022. Louis Vuitton to release new NFTs [Online]. Available: https://www.vogue business.com/technology/louis-vuitton-to-release-new-nfts [Accessed].
Vorhies, D.W., Morgan, N.A., 2003. A configuration theory assessment of marketing organization fit with business strategy and its relationship with marketing performance. J. Market. 67, 100–115.
Wang, Q., Li, R., Wang, Q., Chen, S., 2021. Non-fungible token (NFT): overview, evaluation, opportunities and challenges. arXiv preprint arXiv:2105.07447.
Ward JR., J.H., 1963. Hierarchical grouping to optimize an objective function. J. Am. Stat. Assoc. 58, 236–244.
Wohlin, C., 2014. Guidelines for snowballing in systematic literature studies and a replication in software engineering. Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering, pp. 1–10.
Wu, C.-H., Dong, T.-P., Liu, C.-H., Vu, H.T., 2024. Discovering the critical factors of affecting non-fungible tokens (NFT) purchase intention. Int. Rev. Econ. Finance 96, 103660.
Xie, Q., Muralidharan, S., 2023. It's a comparison game! The roles of social comparison, perceived exclusivity and perceived financial benefits in non-fungible token marketing. J. Res. Indian Med. 18, 294–314.
Xie, Q., Muralidharan, S., Edwards, S.M., 2024. Who will buy the idea of non-fungible token (NFT) marketing? Understanding consumers' psychological tendencies and value perceptions of branded NFTs. Int. J. Advert. 43, 987–1015.
Xie, Q., Muralidharan, S., Edwards, S.M., La Ferle, C., 2023. Unlocking the power of non-fungible token (NFT) marketing: how NFT perceptions foster brand loyalty and purchase intention among millennials and Gen-Z. J. Interact. Advert. 23, 356–373.
Yilmaz, T., Sagfossen, S., Velasco, C., 2023. What makes NFTs valuable to consumers? Perceived value drivers associated with NFTs liking, purchasing, and holding. J. Bus. Res. 165, 114056.
Yin, R.K., 2004. The Case Study Anthology. Sage.
Yoo, K., Welden, R., Hewett, K., Haenlein, M., 2023. The merchants of meta: a research agenda to understand the future of retailing in the metaverse. J. Retailing 99, 173–192.
Zhang, J., Li, J., 2023. A study on consumer perception of NFT based on brand concept map. Europ. J. Inform. Tech. Comput. Sci. 3, 30–34.
Zhang, L., Phang, I.G., 2024. Building brand attachment in China's luxury fashion industry: the role of NFT characteristics. J. Glob. Fashion Market. 1–20.